AI NUMBER PLATE QUALITY ANALYSIS

Computer-Vision Prototype for Parking Image Quality Checks

1. PROJECT OVERVIEW

AI Number Plate Quality Analysis is an independent portfolio project that analyses uploaded vehicle or parking images. It checks image quality, attempts to detect number-plate regions, extracts plate text using OCR, and stores results for later review. The project demonstrates a computer-vision workflow.

2. PROBLEM STATEMENT

Number-plate recognition can be affected by blurry images, poor lighting, low contrast, and plates that are too small or unclear. Manually reviewing every image can also be time-consuming.

This project combines image-quality checks, number-plate detection, and text recognition in one application. It helps identify potentially problematic images and provides results that an operator can verify.

3. HOW IT WORKS

1. **Image Upload:** The user uploads a vehicle image or a close-up of a number plate through the web interface.
2. **Image Quality Analysis:** OpenCV and NumPy calculate blur, brightness, and contrast measurements. Simple thresholds flag images that may require review.
3. **Number-Plate Detection:** A YOLO model attempts to locate the plate and provides a detection confidence score.
4. **Text Extraction:** EasyOCR reads the detected plate region and returns recognised text with an OCR confidence score.
5. **Event Storage:** Analysis results are stored in a local SQLite database for event-history review and CSV export.
6. **Operations Copilot:** Gemini LLM integration allows users to ask natural-language questions about stored parking events.

4. TECHNOLOGIES USED

- **Python:** Core application logic and image-processing workflow.
- **Streamlit:** Interactive web interface and result display.
- **OpenCV and NumPy:** Image decoding and quality measurements.
- **YOLO (Ultralytics):** Number-plate region detection.
- **EasyOCR:** Optical Character Recognition for extracting plate text.
- **SQLite:** Local storage of analysis events.
- **Pandas:** Data handling, filtering, and CSV export.
- **SHA-256:** Identification of identical uploaded files.
- **Gemini:** AI-powered Operations Copilot responses.